Recall Green's First Identity

$$\int_\Omega u \underline{\nabla}^2 w \, d\underline{x} = \int_{\partial\Omega} u \frac{\partial w}{\partial \underline{n}} \, d\sigma - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x}$$

$\underline{n}$ = unit normal vector pointing out of domain

$\frac{\partial w}{\partial \underline{n}} = \underline{\nabla} w \cdot \underline{n}$

Comes from identity $u \underline{\nabla}^2 w = \underline{\nabla} \cdot (u \underline{\nabla} w) - \underline{\nabla} u \cdot \underline{\nabla} w$
+ divergence Thm.

Assumptions: Ω regular (domain, bdd, Lipschitz bdry);
$u, w \in C^1(\Omega) \cap C^0(\bar{\Omega})$ $\left| \int_\Omega u \underline{\nabla}^2 w \, dx \right| < \infty$

To get Green's 2nd Identity
start with Green's 1st identity, interchange u + w
and subtract.

Interchange: $\int_\Omega w \underline{\nabla}^2 u \, d\underline{x} = \int_{\partial\Omega} w \frac{\partial u}{\partial \underline{n}} \, d\sigma - \int_\Omega \underline{\nabla} w \cdot \underline{\nabla} u \, d\underline{x}$

$$\boxed{\begin{array}{l} \underline{\text{Green's 2nd Identity}} \\ \int_\Omega (u \underline{\nabla}^2 w - w \underline{\nabla}^2 u) \, d\underline{x} = \int_{\partial\Omega} \left(u \frac{\partial w}{\partial \underline{n}} - w \frac{\partial u}{\partial \underline{n}}\right) d\sigma \end{array}}$$

Need left-hand side (LHS) integral to be convergent (not $\pm\infty$)
2nd identity is not used as often as first.

§6.2 Heat Eqn

ρ = density $\frac{\text{mass}}{\text{volume}}$

steel has high c

c = specific heat = amount of energy it takes to ~~raise~~ raise the temp. by one degree

aluminum has low c

k = conductivity = how fast heat moves through material

low conductivity
glass, rubber
wood
air (insulation)

high conductivity
copper, steel

$$\rho c u_t - \underline{\nabla} \cdot (k \underline{\nabla} u) = 0$$

$u = u(x, y, z, t)$ = temp.

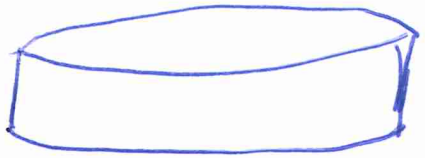
Let $t' = \frac{k}{\rho c} t \Rightarrow \frac{\partial u}{\partial t} = \frac{\partial u}{\partial t'} \frac{dt'}{dt} = \frac{\partial u}{\partial t'} \frac{k}{\rho c}$

$\frac{k}{\rho c}\left[\rho c \left(u_{t'} \frac{k}{\rho c}\right) - \underline{\nabla} \cdot (k \underline{\nabla} u) = 0\right]$ assume k is ~~conductivity~~ constant

$$u_{t'} - \underline{\nabla} \cdot \underline{\nabla} u = 0 \Rightarrow \boxed{u_t - \underline{\nabla} \cdot \underline{\nabla} u = 0}$$

We want to know the temp. $u(\underline{x}, t)$

Cake in oven



We need additional data to solve this problem.

$$u_t - \underbrace{\underline{\nabla} \cdot \underline{\nabla}}\, u = 0$$

$$u_t - \Delta u = 0$$

Initial condition (IC)

$$u(x, y, z, 0) = \underbrace{\phi(x, y, z)}_{\text{must be given}} \quad \text{for all } (x, y, z) \in \Omega$$

Boundary condition: what is happening on $\partial\Omega$? 3 types

1) <u>Dirichlet BC</u> $\quad u(x, y, z; t) = f(x, y, z; t)$ for all $(x, y, z) \in \partial\Omega$, $t > 0$

temp. on bdry is given

i.e. $u = 350°F$ on bdry.

2) Neumann $\frac{\partial u}{\partial \underline{n}}(x,y,z;t) = g(x,y,z;t$

for all $x,y,z \in \partial\Omega$, $t>0$

If there is a perfectly insulated bdry (walls of oven) $\frac{\partial u}{\partial \underline{n}} = 0$

$\underline{\nabla} \cdot \underline{\nabla} u$

$\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle \cdot \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle u$

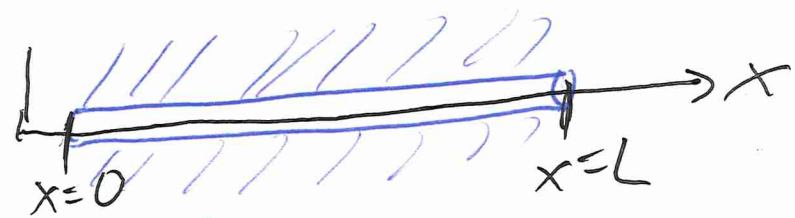
$= u_{xx} + u_{yy} + u_{zz}$

3) Mixed

$\alpha \frac{\partial u}{\partial \underline{n}} + \beta u = h(x,y,z;t)$

Needed if have convective BC

<u>ex</u>) Thin rod, insulated



Find temp.

$\Rightarrow$ assume the temp. does not depend on y, z

$u_t - u_{xx} = 0$

I.C. $u(x,0) = \phi(x)$

B.C. $u(x=0,t) = f(t)$ $\quad u(x=L,t) = g(t)$ $\quad t>0$

$\frac{\partial u}{\partial x}(x=0,t) = \ldots$ $\quad \frac{\partial u}{\partial x}(x=L,t) = \ldots$

also have Dirichlet-Neum.
Neu-Dirich

Ch 6 §3 Laplace Eq'n

$a=1 \quad b=0, \quad c=1$

$b^2-4ac=-4<0$

elliptic $\quad \Delta u = 0 \qquad u_{xx}+u_{yy}+u_{zz}=0 \qquad u_t - \Delta u = 0$

steady state (means the object's temp is not changing anymore) sol'n to the heat eq'n.

No IC needed! (No u_t term)

<u>BC</u> (i) Prescribe temp on bdry (Dirichlet BC)

$$u(x,y,z) = f(x,y,z) \quad (x,y,z) \in \partial\Omega$$

No t

(ii) Prescribe heat flux (Neumann BC)

$$\frac{\partial u}{\partial n}(x,y,z) = g(x,y,z) \quad (x,y,z) \in \partial\Omega$$

No t

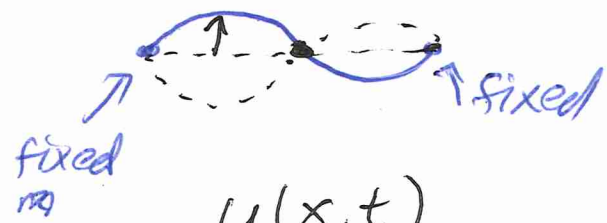
(iii) Mixed BC

$$\alpha(x,y,z)\frac{\partial u}{\partial n}(x,y,z) + \beta(x,y,z)\,u(x,y,z) = h(x,y,z)$$

Note that all quantities are no longer a f'n of time.

ch 6 §4 Wave eq'n

Vibration of a guitar string



$u(x,t)$

u is the displacement from equil.

Assumptions

string vibrates in a plane (true!)
small deformations
density is constant.

$$u_{xx} - \frac{1}{c^2} u_{tt} = 0$$

c is a constant
(a fn of elasticity constant, density)

$$t' = ct$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial t'} \frac{dt'}{dt} = c\, u_{t'}$$

$$\frac{\partial^2 u}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right) = \frac{\partial}{\partial t}\left(c\, u_{t'}\right) = c \frac{\partial u_{t'}}{\partial t'} \frac{dt'}{dt} = c^2 u_{t't'}$$

$$u_{xx} - \frac{1}{c^2}\left(c^2 u_{t't'}\right) = 0 \qquad u_{xx} - u_{tt} = 0$$